

Power Rule & Constant Rule: Finding Derivatives

Calculus Worksheet · Grade 11–12

Name: _____

Date: _____

Learning Objectives

- Apply the Power Rule to find derivatives of polynomial and rational functions
- Apply the Constant Rule to identify the derivative of any constant as zero
- Convert radical and fractional expressions before differentiating using the Power Rule

Problems

1. Use the Constant Rule to find the derivative of the function below.

$$f(x) = 47$$

2. Use the Power Rule to find the derivative of the function below with respect to x .

$$f(x) = x^5$$

3. Use the Power Rule to find the derivative of the function below with respect to x .

$$f(x) = 6x^4$$

4. Find the derivative of the linear function below with respect to x using the Power Rule.

$$f(x) = 9x$$

5. Use the Power Rule and Constant Rule to find the derivative of the polynomial function below with respect to x .

$$f(x) = 4x^3 - 7x^2 + 5x - 12$$

6. Expand the binomial first, then find the derivative of the function below with respect to x .



$$f(x) = (2x + 3)^2$$

7. Convert the radical expression to exponential form, then use the Power Rule to find the derivative with respect to x.

$$f(x) = \sqrt{x}$$

8. Convert the fraction to a negative exponent, then use the Power Rule to differentiate with respect to x. Also apply the Constant Rule where needed.

$$f(x) = \frac{5}{x^3} + e^\pi$$

9. Convert the cube root to exponential form and then find the derivative of the function below with respect to t.

$$g(t) = 4t^5 - \frac{3}{t^2} + \sqrt[3]{t^4}$$

10. Rewrite all terms using exponents, then find the derivative of the function below with respect to x. Simplify and express all answers with positive exponents.

$$h(x) = 3x^6 - \frac{8}{x^4} + 5\sqrt[4]{x^3} - \pi^2 + \frac{1}{2x^5}$$



Power Rule & Constant Rule: Finding Derivatives — Answer Key

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Answer Key

1. Answer: $f'(x) = 0$

- 47 is a constant value — it does not change with respect to x .
- By the Constant Rule, the derivative of any constant is 0.
- Therefore, $f'(x) = 0$.

2. Answer: $f'(x) = 5x^4$

- Apply the Power Rule: if $f(x) = x^n$, then $f'(x) = n \cdot x^{(n-1)}$.
- Here $n = 5$, so bring down the exponent: $f'(x) = 5 \cdot x^{(5-1)}$.
- Simplify: $f'(x) = 5x^4$.

3. Answer: $f'(x) = 24x^3$

- Apply the Power Rule: bring down the exponent as a multiplier.
- $f'(x) = 4 \cdot 6x^{(4-1)} = 24x^3$.
- Therefore, $f'(x) = 24x^3$.

4. Answer: $f'(x) = 9$

- Rewrite as $f(x) = 9x^1$. The exponent is $n = 1$.
- Apply the Power Rule: $f'(x) = 1 \cdot 9x^{(1-1)} = 9x^0$.
- Since $x^0 = 1$, $f'(x) = 9$.

5. Answer: $f'(x) = 12x^2 - 14x + 5$

- Differentiate each term separately.
- Derivative of $4x^3$: $3 \cdot 4x^{(3-1)} = 12x^2$.
- Derivative of $-7x^2$: $2 \cdot (-7)x^{(2-1)} = -14x$.
- Derivative of $5x$: $1 \cdot 5x^{(1-1)} = 5$.
- Derivative of -12 (constant): 0.
- Combine: $f'(x) = 12x^2 - 14x + 5$.

6. Answer: $f'(x) = 8x + 12$

- Expand: $(2x + 3)^2 = (2x + 3)(2x + 3) = 4x^2 + 12x + 9$.
- Differentiate each term: derivative of $4x^2 = 8x$, derivative of $12x = 12$, derivative of $9 = 0$.
- Combine: $f'(x) = 8x + 12$.

7. Answer: $f'(x) = 1 / (2\sqrt{x})$

- Rewrite the radical as an exponent: $f(x) = x^{(1/2)}$.
- Apply the Power Rule: $f'(x) = (1/2) \cdot x^{(1/2 - 1)} = (1/2)x^{(-1/2)}$.
- Rewrite with a positive exponent: $f'(x) = 1 / (2x^{(1/2)}) = 1 / (2\sqrt{x})$



8. Answer: $f'(x) = -15 / x^4$

- Rewrite the fraction: $5/x^3 = 5x^{-3}$. Note that e^π is a constant.
- Differentiate $5x^{-3}$: $(-3) \cdot 5x^{-3-1} = -15x^{-4}$.
- Differentiate e^π using the Constant Rule: derivative = 0.
- Rewrite with a positive exponent: $f'(x) = -15 / x^4$.

9. Answer: $g'(t) = 20t^4 + 6/t^3 + (4/3)t^{1/3}$

- Rewrite each term in exponential form: $4t^5$, $-3t^{-2}$, and $t^{4/3}$.
- Differentiate $4t^5$: $5 \cdot 4t^{5-1} = 20t^4$.
- Differentiate $-3t^{-2}$: $(-2)(-3)t^{-2-1} = 6t^{-3} = 6/t^3$.
- Differentiate $t^{4/3}$: $(4/3)t^{4/3-1} = (4/3)t^{1/3}$.
- Combine: $g'(t) = 20t^4 + 6/t^3 + (4/3)t^{1/3}$.

10. Answer: $h'(x) = 18x^5 + 32/x^5 + (15/4)x^{-1/4} - 5/(2x^6)$

- Rewrite each term: $3x^6$, $-8x^{-4}$, $5x^{3/4}$, $-\pi^2$ (constant), $(1/2)x^{-5}$.
- Differentiate $3x^6$: $6 \cdot 3x^5 = 18x^5$.
- Differentiate $-8x^{-4}$: $(-4)(-8)x^{-4-1} = 32x^{-5} = 32/x^5$.
- Differentiate $5x^{3/4}$: $(3/4) \cdot 5x^{3/4-1} = (15/4)x^{-1/4}$.
- Differentiate $-\pi^2$ using the Constant Rule: derivative = 0.
- Differentiate $(1/2)x^{-5}$: $(-5)(1/2)x^{-5-1} = -(5/2)x^{-6} = -5/(2x^6)$.
- Combine: $h'(x) = 18x^5 + 32/x^5 + (15/4)x^{-1/4} - 5/(2x^6)$.

